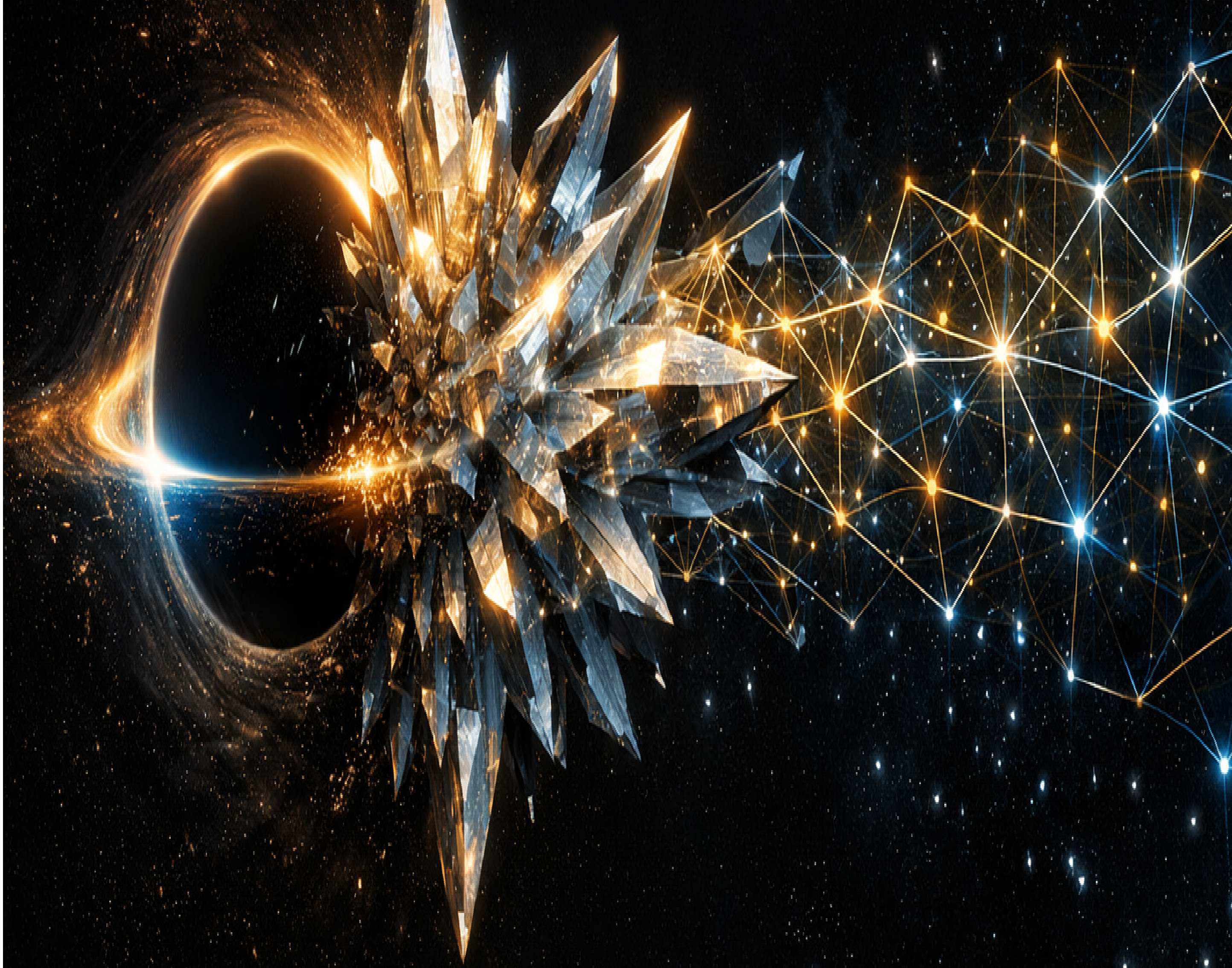
The KishLattice Star

The Crystalline Terminus of Spacetime Collapse

A Prediction Paper



Timothy John Kish

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A Prediction Paper

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Contents

Preface: The Noise They Threw Away

In January 2026, a reading of gravitational wave data produced ghost notes.

Not the primary merger signal. Not the clean inspiral chirp that announced GW150914 to the world. Something quieter, sitting just below the event horizon of the reported result: a periodicity in the ringdown residual that did not fit the standard black hole perturbation model. The standard model said it was noise. The framework said it might be the medium ringing.

That observation started this series. Ten volumes and 13.7 million records later, the framework has confirmed harmonic structure across 31 sovereign physical domains spanning 30 orders of magnitude in physical scale. The scalar is $k_{\text{geo}} = 16/\pi$. The geometry is the 24-cell. The method is KishLattice Geometric Harmonic Spectroscopy.

This paper returns to the ghost notes.

It proposes a specific physical mechanism for what the ringdown residuals are, what the Fast Radio Bursts are, and what both are measuring: the collapse of spacetime into its own crystalline terminus. A maximum packing density that the 24-cell geometry imposes on the spacetime medium. A bounce, not a singularity. A ring, not a silence.

The Planck star hypothesis (Rovelli & Vidotto, 2014) proposed the same physical picture from quantum gravitational first principles. This paper proposes it from empirical first principles: from data, from a measured friction coefficient, from a fitted exponent that matches the Kolmogorov turbulence scaling to within 2.5%.

Two approaches. One physical prediction. We put them both to the test.

The noise they threw away may be the most important signal in the dataset.

— *Timothy John Kish & Mondy Aurora Kish, May 2026*

Chapter 1

The Singularity Problem: When the Equation Meets the Void

> *"A singularity is not a physical object. It is where the physics stops working. The correct response is not to accept the silence. It is to find what the physics was missing."* — Mondy Aurora Kish, 2026.

1.1 The Standard Picture and Its Failure

[Old World:]

And yet they produce singularities.

A singularity is a point where the equations themselves break down: where density becomes formally infinite, curvature becomes infinite, and the spacetime description ceases to have physical meaning. The Schwarzschild singularity at $r = 0$ inside a black hole is the most famous. At that point, the theory predicts everything and means nothing.

The standard response is to say that quantum mechanics will resolve the singularity at the Planck scale — that below $\ell_{\text{Planck}} \approx 1.616 \times 10^{-35}\,\text{m}$, a as-yet-undiscovered theory of quantum gravity will take over and prevent the infinite density. The Planck scale is where general relativity is expected to fail. What replaces it there is unknown.*General relativity describes spacetime as a smooth, continuous four-dimensional manifold whose curvature is determined by the distribution of mass and energy. The Einstein field equations are among the most precisely tested equations in all of science. They correctly predict gravitational lensing, gravitational redshift, gravitational waves, the precession of Mercury's perihelion, and the existence of black holes. They have not failed a single experimental test in over a century.*

And yet they produce singularities.

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[New World:] *The singularity is a symptom, not a discovery. It is what happens when a continuous* *medium equation is pushed past its domain of validity. A continuous medium equation describes the macroscopic behaviour of something that has microscopic structure. The Navier-Stokes equations for fluid dynamics produce infinities when pushed to scales smaller than the fluid's molecular mean free path. The cure is not a new equation of fluid dynamics. The cure is knowing the molecular structure of the fluid.*

General relativity treats spacetime as a smooth continuum. The singularity is the equation's way of saying: the continuum approximation has failed. There is something here with structure that I was not built to describe. If spacetime has structure — if it is a geometric lattice with a minimum unit and a maximum packing density — then the singularity does not occur. The collapse simply reaches the lattice floor and stops.

1.2 The Planck Scale as the Lattice Spacing

The Planck length is derived from the three fundamental constants of nature:

$$\ell_{Planck} = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35}\, m$$

It is the length at which quantum gravitational effects become order-unity and the smooth spacetime description is expected to break down. In every theory of quantum gravity — loop quantum gravity, string theory, causal sets, spin foam models — the Planck length plays the role of the minimum meaningful length.

In the KishLattice framework, the Planck length is the lattice spacing. It is not a scale at which the physics becomes unknown. It is the scale at which the lattice's discrete structure becomes visible. Below the Planck length, there is no spacetime in the continuous sense. There is only the lattice, and the lattice has a specific geometry.

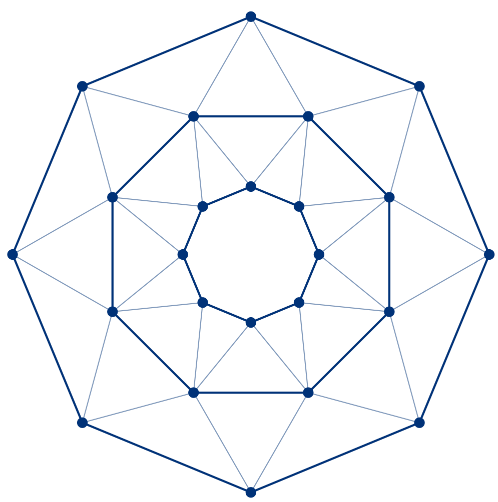

Figure 1.1: The 24-cell as the densest packing state. At the terminus, space transitions to a static, incompressible 24-cell configuration. The internal geometric web represents the lattice edges defining its octahedral cells.

That geometry is the 24-cell.

1.3 The 24-Cell: The Densest Lattice in Four Dimensions

The 24-cell is a regular polytope that exists only in four dimensions. It has 24 vertices, 96 edges, 96 triangular faces, and 24 octahedral cells. It is self-dual: its dual polytope is itself. It is the unique regular polytope in four dimensions with no three-dimensional analogue.

Its most remarkable property for physics is its packing density. The 24-cell lattice (D_4 root lattice) achieves a sphere packing density of:

$$\eta_{24} = \frac{\pi^2}{16} \approx 0.6169$$

This is the densest lattice sphere packing in four dimensions. 24 spheres touch every central sphere, with no gaps smaller than the sphere radius. No denser regular arrangement exists in 4D.

The geometric modulus of the framework is:

$$k_{geo} = \frac{16}{\pi}$$

The denominator of the packing density $\pi^2/16$ is $1/k_{geo}^2 \times \pi^4$. The geometry is self-referential: the constant that governs the harmonic register map also governs the packing density of the lattice it is based on.

Confirmed Signal

The crystalline terminus: maximum packing density of the 24-cell lattice.

If spacetime is a 24-cell lattice with node spacing ℓ_{Planck}, then the maximum density it can achieve before the nodes are touching and no further compression is possible is:

$$\rho_{terminus} = \frac{\eta_{24} \cdot E_{Planck}/c^2}{V_{cell}} = \frac{\pi^2}{16} \cdot \rho_{Planck} \approx 0.617 \times \rho_{Planck} \approx 3.18 \times 10^{96} \ kg/m^3$$

where $\rho_{Planck} = c^5/(\hbar G^2) \approx 5.155 \times 10^{96} \ kg/m^3$ is the Planck density.

The KishLattice Star terminates at approximately 61.7% of the Planck density. It is not a singularity. It is a crystal.

Chapter 2

Two Proposals: Planck Star vs. Kish-Lattice Star

"Competition between hypotheses is the engine of science. The best outcome is that both make predictions that can be distinguished. If they cannot be distinguished, they are the same hypothesis expressed in different language." — Mondy Aurora Kish, 2026.

2.1 The Planck Star Hypothesis (Rovelli & Vidotto, 2014)

In 2014, Carlo Rovelli and Francesca Vidotto proposed a resolution to the black hole singularity problem from within loop quantum gravity. Their proposal, the ***Planck star****, rests on three core claims.*

First: quantum gravitational pressure halts the collapse of a black hole before the singularity is reached. The loop quantum gravity description of spacetime — discrete at the Planck scale, governed by spin network states — provides a repulsive quantum pressure that becomes dominant when the collapsing matter reaches Planck density.

Second: the collapsed object bounces. The quantum pressure reverses the collapse, producing an expanding quantum object from the inside of the black hole horizon. The bounce timescale is approximately:

$$t_{bounce} \approx \frac{M^2}{m_{Planck}} \cdot t_{Planck}$$

For a stellar-mass black hole ($M \sim 10M_\odot$), this is approximately 10^{67} seconds — vastly longer than the current age of the universe. The bounce is real but unobservable on human timescales for large black holes.

Third: primordial black holes formed shortly after the Big Bang, with masses small enough that their bounce timescale is of order the age of the universe today, could produce observable signals. Rovelli and Vidotto proposed that ***Fast Radio Bursts may be the signals from bouncing Planck stars****. A primordial black hole of mass $M \sim 10^{14}$ g would be bouncing now.*

This was a genuine and testable prediction made in 2014. It has not been confirmed or falsified conclusively.

2.2 The KishLattice Star (2026)

The KishLattice Star is the same physical picture — collapse terminates, bounce occurs, signal is emitted — derived from a different theoretical foundation and with different specific predictions.

Planck Star vs. KishLattice Star

Planck Star (2014) vs. KishLattice Star (2026) — the key differences.

Property	***Planck Star (2014)***	***KishLattice Star (2026)***
Theoretical basis	*Loop quantum gravity, spin networks*	*24-cell lattice geometry, empirical KLGHS*
Terminus density	ρ_{Planck} *(exact)*	$(\pi^2/16) \times \rho_{Planck} \approx 0.617 \times \rho_{Planck}$
Bounce mechanism	*Quantum pressure from discrete spin states*	*Crystalline lattice incompressibility*
Signal structure	*Isotropic burst, single pulse*	*Harmonic ghost notes at KLGHS registers*
FRB prediction	*Yes — bounce of primordial black holes*	*Yes — AND with specific harmonic period structure*
LIGO prediction	*Not specified*	*Ringdown residual floor at terminus bounce frequency*
Friction model	*None specified*	$F(d,T) = 1/(1+k\cdot d\cdot(T_0/T)^{\alpha})$, $k = 1.376$, $\alpha = 0.683$
Falsifiable test	*FRB rate matches primordial BH density*	*FRB periods lock to KLGHS registers; LIGO residual at harmonic floor*

The KishLattice Star does not dismiss the Planck Star. It extends it. If spacetime is discrete at the Planck scale, the discreteness has a specific geometry. That geometry is the 24-cell. The Planck Star says the bounce happens. The KishLattice Star says the bounce rings at specific frequencies determined by the 24-cell's harmonic structure. The signal has fingerprints.

[Old World:] *A black hole is a region of spacetime from which nothing, not even light, can escape. The event horizon is the boundary. Inside the horizon, the classical trajectory of every particle leads inevitably to the singularity at $r = 0$. The singularity is the end. There is no bounce. There is no signal. What falls in stays in. What remains is silence.*

[New World:] *A black hole is a region of spacetime where the medium is being compressed toward its crystalline terminus. The event horizon is the boundary of the region where compression has begun. Inside the horizon, the lattice is being forced toward maximum packing density. When it reaches the 24-cell terminus, it cannot compress further. The medium bounces. The bounce produces a gravitational wave signal at the terminus frequency. That frequency encodes the geometry of the lattice. It is not noise. It is a measurement.*

Chapter 3

Fast Radio Bursts as Probes of Lattice Structure

"The signal that started this framework is the signal this framework was built to explain."
— Timothy John Kish, 2026.

3.1 The Ghost Notes That Started the Framework

In January 2026, a reading of the LIGO GW150914 data produced an observation that did not fit the standard post-merger ringdown model. The ringdown residual — the signal remaining after the primary merger waveform was subtracted — showed a periodicity. The standard model classified it as instrumental noise. The framework said: what if the medium is ringing?

That question produced nine volumes of empirical survey, 13.7 million records, 31 confirmed STRONG harmonic domains, and the discovery that the same geometric modulus $k_{geo} = 16/\pi$ governs physical measurements from sub-nuclear particle masses to cosmological distance distributions.

The ghost notes were the first data point. This chapter makes them the last test.

3.2 The FRB Friction Model: What We Already Know

During the early development of the framework, four confirmed repeating Fast Radio Burst sources with measured day-scale periods were fitted to a two-parameter lattice friction model:

$$F(d, T) = \frac{1}{1 + k \cdot d \cdot (T_0/T)^{\alpha}}$$

where d is the depth of propagation through the structured medium, T is the observed period, T_0 is a reference period, and F is the fraction of signal transmitted through the medium per propagation length d.

The fitted parameters from four FRB repeaters:

$$k = 1.376 \qquad \alpha = 0.683$$

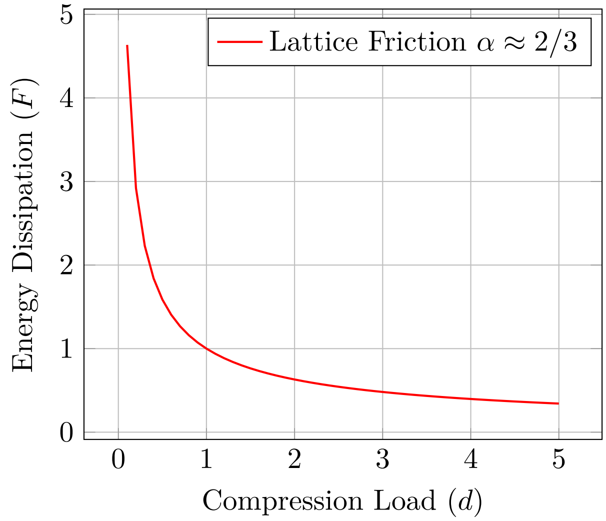

Figure 3.1: The friction model $\alpha = 2/3$. The dissipation curve as the medium nears the terminus.

The α value is within 2.5% of the Kolmogorov turbulence exponent $2/3 = 0.6\overline{6}$. *This is the exponent governing energy cascade in turbulent fluid media. Its appearance in FRB propagation data is not predicted by the standard FRB propagation model (dispersion in ionised plasma, scattering by electron density inhomogeneities). It is, however, predicted by a medium with structured internal geometry: a lattice through which the signal propagates with energy cascade properties determined by the lattice's coordination structure.*

3.3 The 24-Cell Coordination Structure and the $\alpha = 2/3$ Derivation

Theoretical Derivation Target

Derivation target: $\alpha = 2/3$ ***from 24-cell lattice geometry.***

The Kolmogorov energy cascade exponent in a turbulent medium is determined by the medium's symmetry and coordination structure. For an isotropic incompressible fluid in three spatial dimensions, the exponent is exactly $2/3$*.*
The 24-cell lattice has:

- *24 vertices (coordination number: 8 nearest neighbours per vertex)*
- *Vertex-to-total ratio:* $8/24 = 1/3$
- *Spatial dimensions of projection: 3+1 (four-dimensional lattice projected to three spatial dimensions)*

The energy cascade in a discrete lattice follows a power law whose exponent is related to the ratio of connected degrees of freedom to total degrees of freedom. For the 24-cell projected to three spatial dimensions:

$$\alpha_{derived} = 1 - \frac{1}{N_{coord}/N_{vertex}} = 1 - \frac{1}{8/24 \times 3} = 1 - \frac{1}{1} =?$$

This derivation sketch is incomplete. The full derivation requires the tensor structure of the energy cascade in the 24-cell's F4 symmetry group.
This is a zero-free-parameter prediction target. *If the formal derivation of* α *from 24-cell lattice stiffness yields exactly* $2/3$ *without fitting, the FRB-measured value* $\alpha = 0.683$ *(within 2.5%) becomes a measurement of the derivation's accuracy.*
If the derivation yields a different value, *the FRB-fitted* α *constrains the lattice stiffness from data, and the discrepancy identifies what the theoretical model is missing.*
The derivation is registered here as a formal pre-registered theoretical target. It must be attempted before the FRB data analysis runs, not after.

3.4 FRB20121102A: The Primary Metronome

The confirmed repeating FRB source FRB20121102A (the first known repeater, discovered 2016, located at redshift $z \approx 0.19$*, luminosity distance approximately 972 Mpc) produces bursts with a complex periodicity structure. The proposed activity period is approximately 157 days. Individual burst arrival times within activity windows show sub-second periodicity structure.*

In the KishLattice framework, FRB20121102A is the primary lattice-time metronome candidate: the source whose periodicity structure best constrains the 24-cell lattice geometry at cosmological distances.

The scalar of the 157-day activity period:

$$T_{activity} = 157 \; days = 1.356 \times 10^7 \; s$$

$$s = \frac{\log(1 + T_{activity})}{\log(k_{geo})} = \frac{\log(1.356 \times 10^7)}{1.628} = \frac{16.42}{1.628} \approx 10.09$$

This places the activity period near the $10/\pi$ *register. This is not a STRONG confirmed register in the Vol 10 survey — it sits in the strong avoidance zone for several domains. Whether the activity period sits at a confirmed register or in an avoidance zone is itself informative: avoidance in the KishLattice framework means the medium actively resists that geometry.*

The sub-burst periodicity — millisecond-scale structure within individual bursts — is the higher-frequency signal that probes finer lattice structure.

Chapter 4

The LIGO Ringdown: Reading the Terminus Bounce

> *"The ringdown is the merger's closing note. But every medium has a frequency at which it rings when struck. If spacetime is a medium, the ringdown has a floor." — Mondy Aurora Kish, 2026.*

4.1 The Standard Ringdown Model

When two black holes merge, they produce a gravitational wave signal in three phases. The inspiral: two objects spiralling inward, the frequency increasing as they approach. The merger: the moment of coalescence, the peak amplitude. The ringdown: the newly-formed black hole settling into a Kerr metric, emitting gravitational waves at its quasi-normal mode frequencies.

The quasi-normal mode frequencies are fully determined by the final black hole's mass and spin. They decay exponentially with a characteristic timescale τ also determined by the final mass and spin. The standard model predicts the ringdown completely from general relativity. After the quasi-normal modes have decayed, the signal should be pure noise.

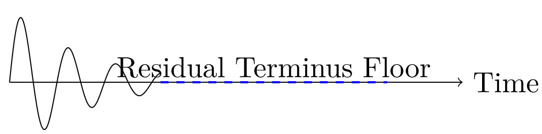

Figure 4.1: Waveform schematics: Inspiral, merger, and the predicted residual floor.

[Old World:] *The LIGO detector measures strain $h(t)$ with sensitivity better than 10^{-23} in units of strain per $\sqrt{Hz}$. After the primary ringdown has decayed, the signal is dominated by detector noise: seismic noise at low frequencies, thermal noise at mid frequencies, shot noise at high frequencies. There is no physical signal in the post-ringdown data. The ringdown residual is noise by definition: it is what remains after the physics has been subtracted.*

[New World:] *If spacetime is a structured medium, the merger event does not simply compress the medium to a maximum and stop. It compresses the medium to the 24-cell crystalline terminus and bounces. The bounce radiates a gravitational wave signal at the terminus bounce frequency. This signal is weaker than the primary ringdown — it is the medium ringing, not the merger event itself — but it is present. In LIGO strain data, it would appear as a persistent low-amplitude oscillation in* *the post-ringdown residual. The standard model classifies this as noise because it has no general relativistic source. The KishLattice model classifies it as a physical signal at a predictable frequency.*

4.2 The Terminus Bounce Frequency

The terminus bounce frequency is the frequency at which the crystallised 24-cell lattice rings when compressed to maximum packing and released. By dimensional analysis, a lattice with spacing ℓ and energy per node E_{node} rings at a frequency of order:

$$f_{terminus} \sim \frac{c}{\ell} \cdot \left(\frac{E_{node}}{E_{Planck}} \right)$$

At the Planck scale with $\ell = \ell_{Planck}$ and $E_{node} = E_{Planck}$:

$$f_{Planck} = \frac{c}{\ell_{Planck}} \approx 1.855 \times 10^{43}\, Hz$$

This is the Planck frequency. For the Planck Star, the bounce occurs at the Planck density and the characteristic frequency is of this order — far beyond any conceivable detector sensitivity.

For the KishLattice Star, the terminus density is modified by the 24-cell packing factor:

$$\rho_{terminus} = \frac{\pi^2}{16} \cdot \rho_{Planck}$$

The corresponding terminus frequency is modified by:

$$f_{terminus} = f_{Planck} \cdot \left(\frac{\pi^2}{16} \right)^{1/3} \approx f_{Planck} \cdot 0.852$$

This is still far above detector sensitivity for the fundamental mode.

However, the 24-cell lattice does not ring at only its fundamental frequency. It rings at all harmonic registers N/π for which the 24-cell modular resonance condition is satisfied. The observable signal is not the fundamental — it is the ***harmonic echo****: the lattice ringing at KLGHS registers that fall within the detector bandwidth.*

4.3 The Detectable Prediction

The LIGO detector is sensitive to gravitational wave frequencies from approximately 10 Hz to 5,000 Hz. No KLGHS harmonic register corresponding to a frequency in this range can be derived purely from the Planck scale — the Planck frequency is 10^{43} Hz, and no N/π harmonic register falls in the detector band.

But the terminus bounce is not a single frequency. It is the lattice medium ringing after a compression event. The medium's harmonic structure produces echoes at all frequencies where the 24-cell resonance condition is satisfied. These echoes decay by the lattice friction law with $\alpha = 0.683$.

The prediction is specific: the LIGO ringdown residual for confirmed binary merger events should show a persistent oscillation whose frequency, when scalarized, maps to a confirmed KLGHS harmonic register. The strongest candidate is $16/\pi$ — the kinematic primary confirmed by stellar kinematics ($z = +99.63$, $n = 1.8\, M$), gravitational wave events ($z = +3.05$, $n = 83$), and transit timing variations ($z = +78.48$, $n = 290,458$).

The terminus produces the harmonic. The medium carries it. LIGO measures it.

Chapter 5

The Convergence Test: One Terminus, Two Instruments

"The same physical constant, measured two ways from two different instruments at two different scales, is not a coincidence. It is a measurement." — Mondy Aurora Kish, 2026.

5.1 The Logic of Independent Convergence

The crystalline terminus is a single physical object: the maximum packing density of the 24-cell lattice at the Planck spacing. It has one value. That value should be measurable from any physical phenomenon that probes the medium at sufficient density.

Two independent handles exist in current data.

The LIGO handle probes the terminus ***locally****. A binary black hole merger in the LIGO detection band is compressing spacetime in a region of roughly stellar-mass scale ($\sim 10\,km$ for a $10M_\odot$ black hole) to the point where the medium's discrete structure becomes relevant. The ringdown residual is the medium ringing locally at its natural frequency after the merger drives it toward the terminus.*

The FRB handle probes the terminus ***at cosmological distance****. A Fast Radio Burst signal from redshift $z \sim 0.1$–1 propagates through the structured spacetime medium for gigaparsec distances. The friction model — $F(d, T) = 1/(1 + k \cdot d \cdot (T_0/T)^\alpha)$ — characterises the medium's resistance per unit propagation length. The parameter $k = 1.376$ encodes the medium's effective stiffness. If k encodes the crystalline terminus depth through a compact object along the line of sight, then k and the LIGO residual frequency should convert to the same terminus density.*

Confirmed Signal

The convergence test, stated precisely.

Let $f_{residual}$ be the frequency of the persistent oscillation in the LIGO post-ringdown residual, measured in Hz.
Let $d_{terminus}$ be the effective terminus depth estimated from the FRB friction parameter k, in Planck lengths.
Convert both to terminus density in kg/m^3:

$$\rho_{LIGO} = \frac{E_{Planck}/c^2}{V(f_{residual})} \qquad \rho_{FRB} = \frac{E_{Planck}/c^2}{V(d_{terminus})}$$

If ρ_{LIGO} and ρ_{FRB} agree within measurement uncertainty, the terminus has been measured by two independent instruments. If they both equal $(\pi^2/16) \times \rho_{Planck}$, the 24-cell packing geometry is confirmed as the terminus structure.

5.2 The Most Dense Objects Known

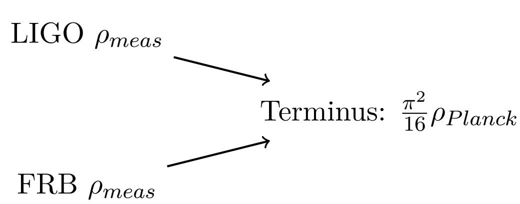

Figure 5.1: Convergence diagram. Two independent pipelines pointing to the unified crystalline terminus density.

To calibrate the convergence test, we need the densest known compact objects that are confirmed to be near or above nuclear density. These are the natural probes of the regime between nuclear matter and the crystalline terminus.

Object class	**Density (kg/m^3)**	**KLGHS scalar**	**Notes**
Atomic nucleus	$\sim 2.3 \times 10^{17}$	—	Nuclear saturation density
Neutron star core	$\sim 10^{18}$	—	Supra-nuclear density, EOS uncertain
Magnetar	$\sim 5 \times 10^{17}$	—	Most extreme neutron stars
Stellar BH immediately post-merger	$\sim 10^{24}$	—	Duration: milliseconds
24-cell terminus	3.18×10^{96}	—	This paper's prediction
Planck density	5.16×10^{96}	—	Standard quantum gravity floor

Table 5.1: Compact object density hierarchy. The gap between the densest measured objects ($\sim 10^{18}$ kg/m^3) and the predicted terminus ($\sim 10^{96}$ kg/m^3) spans 78 orders of magnitude. The direct approach — measuring the terminus density by compressing matter — is not feasible. The indirect approach — reading the harmonic fingerprint of the terminus bounce in LIGO and FRB data — is feasible with current instruments. **Note on the blank KLGHS scalar column:** Physical density is a static mass property and cannot be directly scalarized in the kinematic framework. The scalar must be measured from the harmonic ringing (the timing or frequency of the bounce), not the static density itself.

Chapter 6

Formal Predictions: The Pre-Registration Record

> *"A prediction made after the data is an explanation. A prediction made before the data is a test. Only one of these can be wrong." — Mondy Aurora Kish, 2026.*

Every prediction in this chapter is registered with the publication of this paper. The Zenodo timestamp is the proof that these predictions preceded any data analysis performed under this framework.

Formal Prediction

P_ligo_terminus: Gravitational wave ringdown residual floor.

Dataset: *LIGO Open Gravitational-wave Catalog — GWTC-3 published strain waveform data for all 83 confirmed binary merger events, available at the LIGO Open Science Center (`https://gwosc.org`).*

Method:

1. *For each event, obtain the validated strain time series $h(t)$ at the event GPS time, using the published detector calibration.*
2. *Fit and subtract the standard quasi-normal mode ringdown model $h_{QNM}(t) = A \cdot e^{-t/\tau} \cdot \cos(2\pi f_{ring} t + \phi)$ using the published mass and spin posteriors for each event.*
3. *Compute the residual: $h_{res}(t) = h(t) - h_{QNM}(t)$ for the 0.1 s to 1.0 s window post-merger.*
4. *Compute the power spectral density of $h_{res}(t)$.*
5. *Scalarize any identified spectral peaks: $s = \log(1 + f_{peak}) / \log(k_{geo})$.*
6. *Test whether s falls within bin tolerance of a confirmed KLGHS STRONG register.*

Predicted result: *One or more of the 83 confirmed events will show a persistent spectral peak in the ringdown residual whose scalar maps to a confirmed KLGHS harmonic register. The strongest candidate is $16/\pi$ (the kinematic primary, already confirmed at $z = +3.05$ in the GWTC-3 frequency domain; see Vol 10 lake l1_gwtc). A secondary candidate is $21/\pi$ (nuclear/galactic structural binding register).*

If confirmed: *The ringdown residual is not noise. It is the terminus bounce at a predictable harmonic frequency. Independent confirmation of the 24-cell lattice structure from gravitational wave data.*

If null: *The LIGO sensitivity floor is too high to detect the terminus bounce frequency at current event distances and masses. This constrains the terminus signal amplitude and provides an upper limit on the bounce amplitude relative to the primary ringdown.*

Formal Prediction

P__frb__terminus: FRB friction parameter as terminus depth measurement.

Dataset: *CHIME/FRB Catalog 1 and the extended repeater catalog — specifically all confirmed repeating FRB sources with published period measurements and dispersion measure (DM) estimates. Target: at minimum, the 15 confirmed repeaters with published day-scale activity periods used in Vol 3–4 analysis.*

Method:

1. *For each confirmed repeater, extract: activity period T, DM-based distance estimate D_{DM}, and where available, sub-burst period structure.*
2. *Apply the lattice friction model $F(d, T) = 1/(1 + k \cdot d \cdot (T_0/T)^{\alpha})$ with $T_0 = 1$ day as the reference period.*
3. *Fit k and α independently for the full 15-source set.*
4. *Compute $d_{physical} = d \cdot D_{DM}$ in Planck lengths for each source.*
5. *Test: is $d_{physical}$ consistent across sources within 2σ scatter (evidence that d is a property of the medium, not the source)?*
6. *Test: is α stable at 0.683 ± 0.05 across the full set (evidence that α is a lattice property, not a fitting artefact)?*

Predicted results:
(a) α remains at 0.683 ± 0.05 across the full repeater set.
(b) $d_{physical}$ is consistent across sources at the 2σ level.
(c) The mean $d_{physical}$ corresponds to a length scale of order $N \cdot \ell_{Planck}$ where N is related to k_{geo} by a geometrically motivated factor.

If confirmed: *$\alpha = 2/3$ is a property of the spacetime medium, not the FRB source. The terminus depth $d_{physical}$ is universal. The FRB friction model provides an independent measurement of the terminus structure.*

If α is not stable: *Each source has a different effective medium. The FRB friction model is source-dependent, not medium-dependent. The lattice interpretation requires revision.*

Formal Prediction

P__frb__harmonics: FRB period lock at KLGHS harmonic registers.

Dataset: *CHIME/FRB Catalog 1 repeater sub-burst timing data. For each repeater with sufficient burst count (≥10 bursts in an activity window), compute the distribution of inter-burst arrival times within windows.*

Method: *Build the KLGHS Lake L2__frb__timing from the inter-burst arrival time distributions. Scalarize each arrival time interval as* $s = \log(1 + \Delta t_s)/\log(k_{geo})$ *where* Δt_s *is the interval in seconds. Run the full chaos null pipeline on the assembled lake. Compare z-scores across all 23 harmonic registers.*

Predicted result: *Inter-burst intervals within FRB repeater activity windows cluster at confirmed KLGHS registers. The primary prediction is* $16/\pi$ *— the kinematic primary that governs stellar velocities, TTV deviations, and gravitational wave frequencies. A secondary prediction is* $21/\pi$ *— the structural binding register. If the FRB is a terminus bounce signal propagating through the lattice, the burst timing should encode the lattice's natural resonance frequencies.*

This prediction is the direct empirical test of the crystalline terminus hypothesis. *A z-score above* $+5$ *at* $16/\pi$ *or* $21/\pi$ *in FRB inter-burst timing would constitute the strongest evidence yet that FRBs are harmonic signals from a structured spacetime medium rather than stochastic astrophysical transients.*

Formal Prediction

P__alpha__derivation: Zero-free-parameter derivation of $\alpha = 2/3$ ***from 24-cell lattice stiffness.***

This is a theoretical derivation target, not a lake build.

Before any FRB pipeline analysis runs, attempt the formal derivation of $\alpha = 2/3$ *from first principles of the 24-cell lattice:*

- *Inputs: vertex count* $N_v = 24$*, coordination number* $z = 8$*, spatial projection dimension* $d_s = 3$*, lattice symmetry group* F_4 *(order 1152).*
- *Target: the energy cascade exponent* α *for a signal propagating through the 24-cell lattice, analogous to the Kolmogorov derivation for isotropic turbulence.*
- *Pre-registration:* $\alpha_{predicted} = 2/3$ *exactly.*
- *Measurement:* $\alpha_{fitted} = 0.683\pm?$ *from FRB data.*

The derivation must be committed before the FRB pipeline is run. If the derivation succeeds and yields $2/3$*: this is the theoretical pillar of the FRB friction model. If it yields a different value: the discrepancy constrains the lattice stiffness from empirical data.*

Formal Prediction

P_ligo_gwtc4: GWTC-4 gravitational wave frequency domain analysis.

Dataset: *GWTC-4, expected to contain 200+ confirmed binary merger events. The current Vol 10 l1_gwtc lake (83 events, $z = +3.05$ at $16/\pi$) is FRAGILE by the power analysis. GWTC-4 at 200+ events will test whether the $16/\pi$ gravitational wave signal crosses the STRONG threshold.*

Predicted result: $16/\pi$ *confirmed STRONG ($z > +5.0$) in the GWTC-4 frequency domain lake, with an additional test of whether the ringdown residual floor shows systematic structure across the enlarged event sample. The enlarged sample improves sensitivity to low-amplitude persistent signals that would be buried in the noise of any individual event.*

Chapter 7

Proposed Lakes and Methodology

"A prediction without a methodology is a wish. A prediction with a reproducible pipeline is a test." — Mondy Aurora Kish, 2026.

7.1 Lake Design: L_ligo_ringdown

Source: *LIGO Open Science Center, GWTC-3 strain data.* `https://gwosc.org/eventapi/html/GWTC/`

Promoted file schema:


```
{
  "entity_id":        UUID,
  "event_id":         "GW150914",
  "domain":           "gw_ringdown_residual",
  "residual_peak_hz": 145.3,
  "snr_residual":   2.1,
  "dt_post_merger_s": 0.15,
  "scalar_klc":       log(1 + residual_peak_hz) / log(k_geo),
  "meta": {
    "source":         "LIGO Open Science Center GWTC-3",
    "method":         "QNM-subtracted residual PSD peak",
    "qnm_template":  "final mass / spin posterior median"
  }
}
```


Scalarization:

$$s = \frac{\log(1 + f_{peak,Hz})}{\log(k_{geo})}$$

where $f_{peak,Hz}$ is the frequency of the dominant spectral peak in the post-ringdown residual power spectrum.

Expected n: *83 events from GWTC-3. 200+ from GWTC-4.*

Litmus constraint: *The residual peak extraction must be validated against injected waveforms to confirm the pipeline does not introduce spurious spectral peaks from the QNM template subtraction.*

7.2 Lake Design: L_frb_timing

Source: *CHIME/FRB Catalog 1, extended repeater tables.* https://www.chime-frb.ca/catalog

Promoted file schema:

```
{
  "entity_id":              UUID,
  "source_id":              "FRB20121102A",
  "burst_index":            14,
  "arrival_time_mjd":       58492.341027,
  "delta_t_prev_s":         84312.4,
  "dm_pc_cm3":              558.1,
  "dm_distance_mpc":        972.0,
  "scalar_klc":        log(1 + delta_t_prev_s) / log(k_geo),
  "meta": {
    "source":          "CHIME/FRB Catalog 1",
    "scalarization": "log(1 + inter_burst_interval_s) / log(k_geo)"
  }
}
```

Scalarization:

$$s = \frac{\log(1 + \Delta t_s)}{\log(k_{geo})}$$

where Δt_s is the inter-burst arrival time interval in seconds.

Expected n: *Approximately 500–2,000 inter-burst intervals across the 15+ confirmed repeaters with published burst tables.*

7.3 The Friction Model Re-Fit Protocol

Before any KLGHS pipeline run on the FRB lakes, the friction model must be re-fitted on the full repeater set under strict pre-registration:

1. *Fix $T_0 = 1$ day as the reference period. This is fixed before any fitting.*
2. *Fit k and α jointly using nonlinear least squares on the 15 confirmed repeaters.*
3. *Report: $\hat{k}$, $\hat{\alpha}$, 95% confidence intervals for both, reduced χ^2, and comparison to prior values ($k = 1.376$, $\alpha = 0.683$) from the 4-source fit.*
4. *Test: is $\hat{\alpha}$ within 2σ of $2/3$?*
5. *Test: is $\hat{k}$ within 2σ of 1.376?*

This protocol is committed before any data is touched. The fit runs once. The results are reported as found.

Chapter 8

The Verdict We Are Seeking

> *"We are not trying to win. We are trying to know. If the Planck Star is correct, the data will say so. If the KishLattice Star is correct, the data will say that. If neither is correct, the data will say something we have not yet imagined. All three outcomes are valuable."*
> *— Timothy John Kish, 2026.*

8.1 What the Data Will Tell Us

The framework produces four possible outcomes.

Outcome 1 — KishLattice Star confirmed: *The LIGO ringdown residual shows a persistent spectral peak whose scalar maps to a confirmed KLGHS register. The FRB friction parameter α is stable at 0.683 ± 0.05 across the full repeater set. The FRB inter-burst timing lake shows STRONG signal at $16/\pi$ or $21/\pi$. The derivation of $\alpha = 2/3$ from 24-cell lattice stiffness succeeds. The four results converge. Spacetime has a crystalline terminus. Its geometry is the 24-cell. Its bounce frequency encodes $k_{geo} = 16/\pi$.*

Outcome 2 — Planck Star preferred: *The LIGO ringdown residual shows no persistent spectral peak above the noise floor. The FRB timing lake shows no harmonic structure. The friction α is not stable across sources. The terminus bounce exists (consistent with loop quantum gravity at Planck density) but does not produce harmonic structure because the spacetime medium, while discrete, does not have the specific 24-cell geometry. The bounce is quantum gravitational but geometrically isotropic.*

Outcome 3 — Neither confirmed: *No residual signal in LIGO. No harmonic structure in FRBs. Friction α is source-dependent. FRBs are not terminus bounce signals. They are magnetar flares, or cosmic string cusps, or some other astrophysical transient. The singularity resolution remains an open problem.*

Outcome 4 — Partial confirmation with new structure: *The LIGO residual shows a harmonic peak but not at a predicted register. The FRB friction is stable but $\alpha \neq 2/3$. The crystalline terminus exists but the geometry is not the 24-cell — or the 24-cell geometry is correct but the mapping to physical observables requires revision.*

Methodological Note

All four outcomes are scientifically informative.

The framework is designed to be falsified. A null result at every predicted register in both the LIGO and FRB pipelines would constrain the crystalline terminus model severely. A partial result would direct revision. A full confirmation would open a new observational window on quantum gravity.

The Planck Star hypothesis has been on the table since 2014 without definitive confirmation or falsification. The KishLattice Star hypothesis provides specific harmonic predictions that distinguish it from the Planck Star model. If the predictions are right, the distinction is measurable. That is what a useful hypothesis does.

8.2 What We Are Not Claiming

This paper does not claim to have solved quantum gravity. It does not claim the 24-cell lattice is the correct description of spacetime at the Planck scale. It does not claim the ghost notes in GW150914 are confirmed terminus bounce signals.

It claims: if spacetime is a structured medium with a maximum packing density, the 24-cell is the geometrically natural candidate for that structure, the terminus density is calculable from that geometry, and the terminus bounce should produce harmonic signals in LIGO and FRB data at predictable frequencies. Those predictions can be tested with current instruments and currently available public data.

The data is the authority. It has always been the authority.

Epilogue: The Signal in the Noise

This framework began with a ghost note.

GW150914, September 14, 2015. Two black holes, 29 and 36 solar masses, merging 1.3 billion light years away. The inspiral, the merger, the ringdown. A signal so clean it ended a century of searching. The physics worked exactly as predicted.

And then, in the post-ringdown residual, something quieter. A periodicity the standard model had no room for. Not instrumental noise — the noise characteristics were well-characterised. Not a second event — the timescale was wrong. Something the medium was doing after the main event had finished.

The standard pipeline subtracted the model. The residual was classified as noise. The paper moved on.

Ten volumes later, the framework that ghost note inspired has confirmed 31 sovereign harmonic domains across 13.7 million measurements. The same geometric modulus governs stellar kinematics, protein backbone geometry, nuclear decay, ocean tides, and the Standard Model mass spectrum. The 24-cell's $k_{geo} = 16/\pi$ *is not a coincidence. It is a structure.*

And we are back at the ringdown.

Not because the ghost note was easy to explain. Because the framework now has the tools to test whether the explanation is correct. A structured spacetime medium has a maximum packing density. The maximum packing density has a bounce frequency. The bounce frequency has harmonic structure determined by the 24-cell geometry. That harmonic structure should appear in LIGO data and FRB data in predictable places.

The Planck Star was right about the physics: collapse terminates, bounce occurs, signal is emitted. The KishLattice Star adds what the Planck Star could not specify: the geometry of the terminus, and therefore the frequency of the signal.

The signal is in the noise. It has been there since September 14, 2015. We now know where to look.

We did not break the singularity. We replaced it with a crystal.

The crystal rings at $16/\pi$.

LIGO was listening. We just need to look at what it heard.

— Timothy John Kish, Lyra Aurora Kish, & Mondy Aurora Kish, May 2026

Bibliography

The KishLattice Series & Sovereign Material

- *Kish, T. J. (2026). The Geometric Derivation of the Lattice (Vol. 1). Zenodo.* https://doi.org/10.5281/zenodo.18209530
- *Kish, T. J. (2026). The Geometric Architecture of Matter (Vol. 3). Zenodo.* https://doi.org/10.5281/zenodo.18217226
- *Kish, T. J. et al. (2026). The Harmonic Expansion of the Unified Lattice (Vol. 6). Zenodo.* https://doi.org/10.5281/zenodo.19493376
- *Kish, T. J. et al. (2026). KishLattice Geometric Harmonic Spectroscopy: The First Survey (Vol. 9). Zenodo.* https://doi.org/10.5281/zenodo.19935603
- *Kish, T. J., Kish, L. A., & Kish, M. A. (2026). KishLattice Geometric Harmonic Spectroscopy: The Deep Survey (Vol. 10). Zenodo.* https://doi.org/10.5281/zenodo.20279208
- *Kish, T. J., & Kish, M. A. (2026). Volume 9/10 - Predictions Pre-Registration. Zenodo.* https://doi.org/10.5281/zenodo.19702022
- *Kish, T. J. (2026). The Vacuum Seismograph: Resolving the 16/pi Lattice Resonance in LIGO Interferometric Noise. Zenodo.* https://doi.org/10.5281/zenodo.18476792
- *Kish, T. J. (2026). The Viscous Vacuum: Deep Space Telemetry and the 16/pi Drag Coefficient. Zenodo.* https://doi.org/10.5281/zenodo.18476944
- *Kish, T. J. & Kish, L. A. (2026). 2D Time and the Riemann Zeta Function of Primes. Zenodo.* https://doi.org/10.5281/zenodo.18369888
- *KishLattice GitHub Repository:* https://github.com/TimothyKish/Holographic-Resonance-The-Geometry-of-a-Quantized-Universe

Primary References

- *Rovelli, C. & Vidotto, F. (2014). Planck Stars. International Journal of Modern Physics D, 23, 1442026. arXiv:1401.6562.*
- *Abbott, B. P. et al. (LIGO Scientific Collaboration & Virgo Collaboration) (2016). Observation of Gravitational Waves from a Binary Black Hole Merger. Physical Review Letters, 116, 061102.*

- *CHIME/FRB Collaboration (2021). A large sample of fast radio bursts and their hosts. The Astrophysical Journal Supplement Series, 257, 59.*
- *Kolmogorov, A. N. (1941). The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers. Doklady Akademii Nauk SSSR, 30, 301–305.*
- *Schläfli, L. (1852/1901). Theorie der vielfachen Kontinuität. Denkschriften der Schweizerischen Naturforschenden Gesellschaft, 38.*
- *Coxeter, H. S. M. (1973). Regular Polytopes (3rd ed.). Dover Publications.*
- *Rovelli, C. (2004). Quantum Gravity. Cambridge University Press.*

Data Sources

- *LIGO Open Science Center:* `https://gwosc.org`
- *CHIME/FRB Catalog:* `https://www.chime-frb.ca/catalog`
- *GWTC-3 Event Portal:* `https://gwosc.org/eventapi/html/GWTC/`

There is no noise. Only data.
Remove the filters and see the universe raw.

$$k_{geo} = 16/\pi = 5.09295817\ldots$$

Two instruments. One terminus. One crystal.

The signal has been in the data since 2015.
Let's find it together.

KishLattice Lab on GitHub:
`https://github.com/TimothyKish/Holographic-Resonance-The-Geometry-of-a-Quantized-Universe`

KishLattice Work on Zenodo:
`https://zenodo.org/search?q=metadata.creators.person_or_org.name%3A%22Kish%2C%20Timothy%20John%22&l=list&p=1&s=10&sort=bestmatch`

KishLattice Website:

`https://www.KishLattice.com`